

Improving Low-Resolution Face Recognition under Limited Data

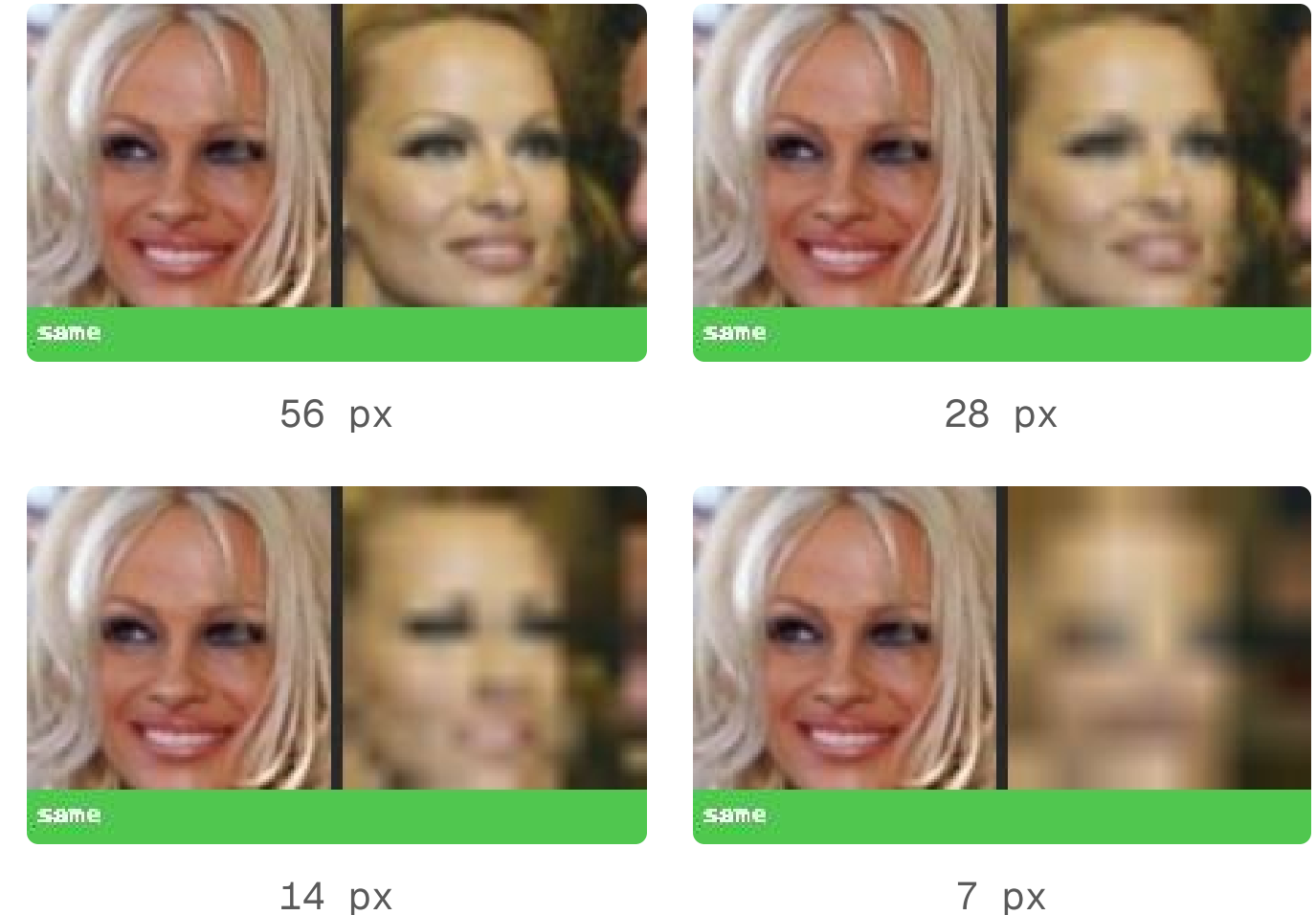
How synthetic data generation can close the domain gap

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HR gallery, **low-resolution** probe

- Watchlist / forensics: enrol a mugshot or ID photo, match a face captured at distance.
- After detection and alignment the face region is often 16–32 px. The recogniser expects 112 px.
- Upsampling does not put identity back. Blur, sensor noise and compression form a domain shift the HR-trained model never saw.



One verification pair, downsampled to the target size and upsampled back to 112 px.

The scarcity is on the **LR** side

LABELLED HR

Abundant

WebFace4M/12M, millions of identities.

LABELLED NATIVE LR

Scarce

Few identities, no control over capture.

PAIRED NATIVE LR / HR

Essentially unavailable

Same subject, same time, both resolutions.

Standard practice: **synthesise** LR by degrading HR faces, train on it, and validate on **synthetically degraded** benchmarks.

Question we set out to answer: **does a method tuned on synthetic LR actually work on real LR** — and how much synthesis effort is repaid in accuracy?

Three levels of synthesis effort

INTERPOLATION

low

Downsample → upsample

- WebFace4M to 56 / 28 / 14 / 7 px, {area, cubic} down × {area, cubic} up, back to 112 px
- Retrain EdgeFace-S end-to-end (CosFace, PartialFC, 100 epochs)
- Deterministic, no learned part

REAL-ESRGAN DEGRADATION

medium

Stochastic sensor model

- Random blur kernels, resize, Gaussian/Poisson noise, JPEG, 2nd-order pass
- Native 32 px copies paired with 112 px HR
- Also a native-32 px stem variant (1×1 conv replaces stride-4 stem)

LEARNED SR FRONT-END

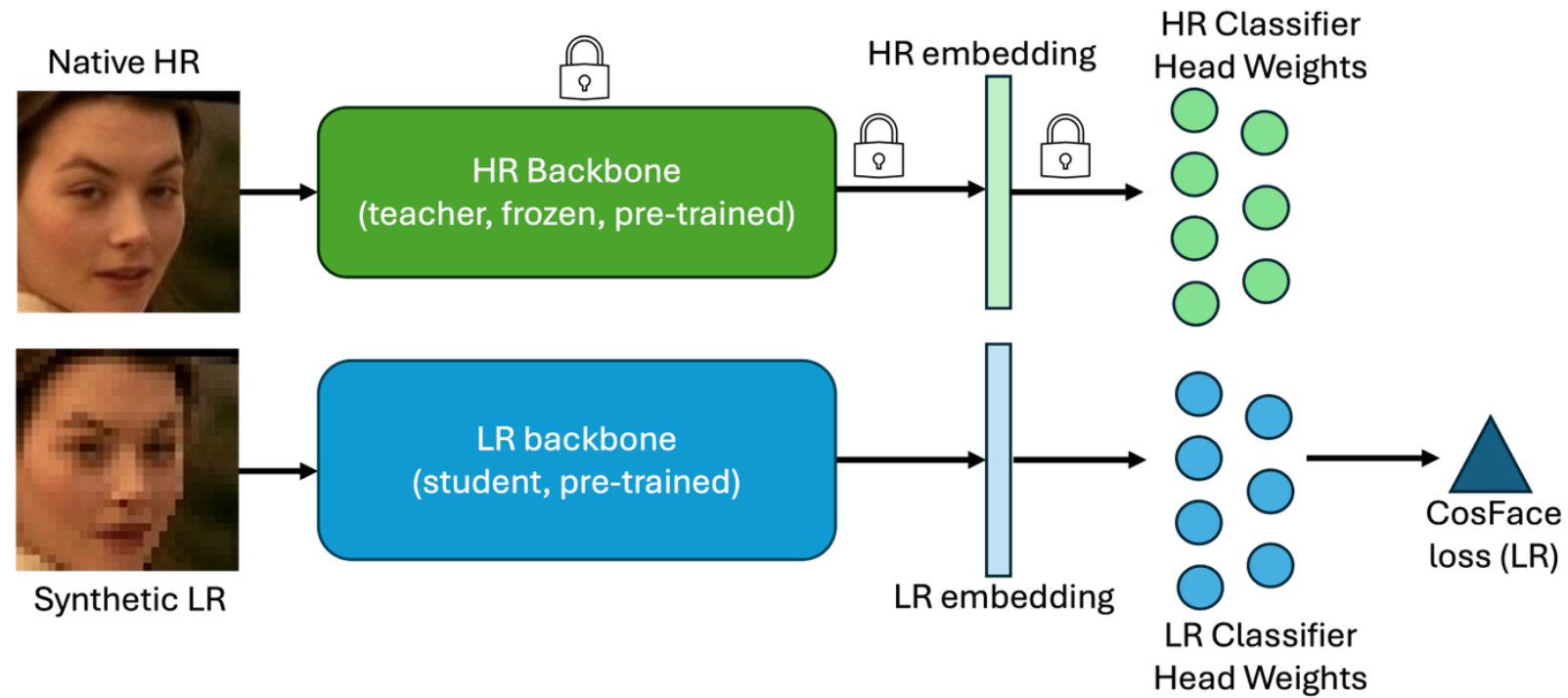
high

Identity-aware super-resolution

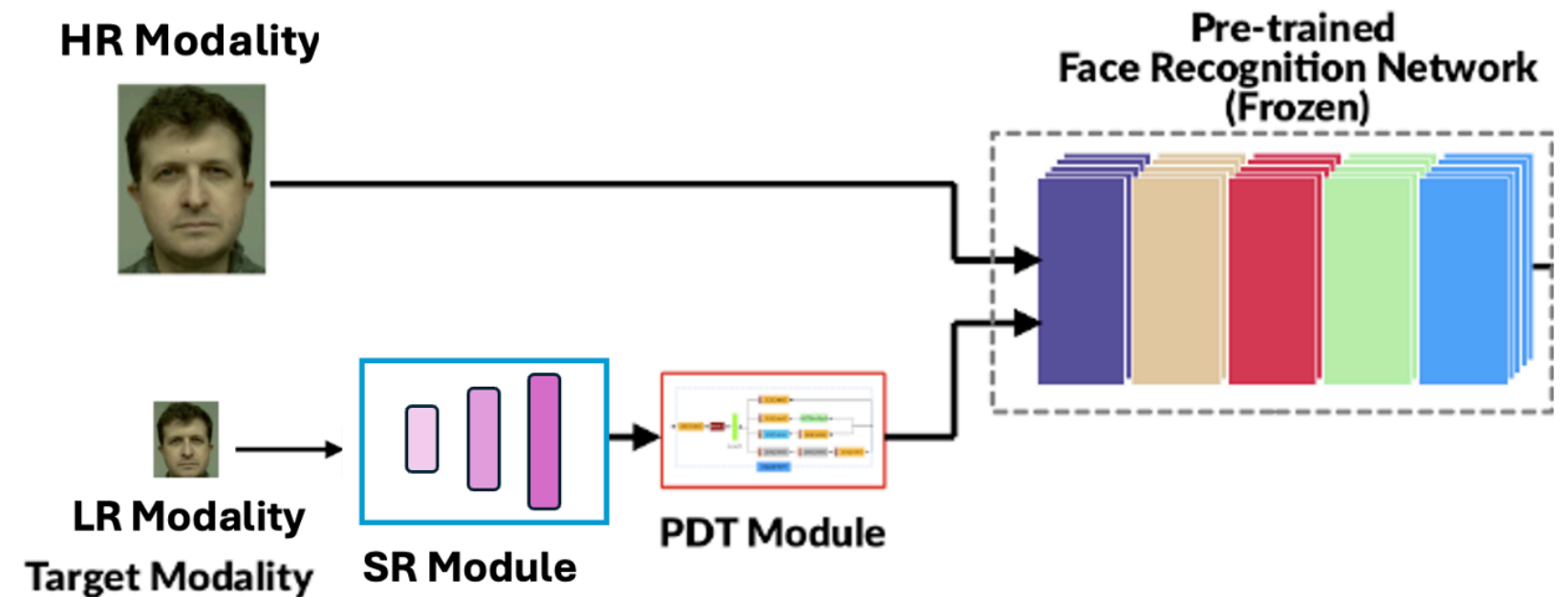
- 32→64→128 px via PixelShuffle, valid convs to 112 px; ESPCN (0.86 M) or RRDB (6.5 M)
- Stage 1: L1 + identity cosine loss with frozen backbone
- Stage 2: SR + PDT trained jointly, contrastive; backbone frozen

Notation: $s \downarrow d / \uparrow u$ = downsample to s px with d , upsample to 112 px with u . $28 \downarrow c / \uparrow a$ = cubic down, area up. Recogniser-side choice (retrain / distil / freeze + adapt input) is varied independently.

Retrain, distil, or freeze and adapt the input



Knowledge distillation. Frozen HR EdgeFace-S teacher; LR student matches its embeddings and PartialFC weights on top of CosFace.



Input-side adaptation. SR upsampler → Prepend Domain Transformer (PDT) → frozen EdgeFace-base (WebFace12M). Only the front-end is trained.

Direct feed is the reference for everything: align the LR face, resize to 112 px, feed the HR-trained backbone. No front-end, no LR training. It turns out to be a hard baseline.

Synthetic benchmarks **and** real native LR

SYNTHETIC CROSS-RESOLUTION

- LFW, CFP-FP, AgeDB-30 degraded to 56 / 28 / 14 / 7 px
- HR→LR: one image degraded (the surveillance case)
- LR→LR: both degraded (harder bound)
- Plus standard HR verification and IJB-C

REAL NATIVE LR

- **TinyFace** identification: mAP, Rank-k; gallery ≈ 157 k
- Two alignment pipelines: `aligned_pad` crops and the CVLface DFA aligner (ResNet-50)
- Every model compared against the **direct feed of its own backbone**

Fairness: RFW (African, Asian, Caucasian, Indian), thresholds calibrated on IJB-C at $\text{TMR@FMR}=10^{-2}$. All numbers are percentages, flip-augmented at test time.

On synthetic LR, 28 ↓c/↑a wins for every method

Macro-mean acc.	HR-tr.	56 ↓c/↑a	28 ↓a/↑c	28 ↓c/↑a	14 ↓a/↑c
HR → LR					
LR backbone	78.85	80.25	78.77	80.74	79.67
KD	78.85	80.74	78.60	82.19	79.82
PDT	78.85	79.86	80.35	81.22	79.92
LR → LR					
LR backbone	78.28	78.92	78.74	79.06	78.31
KD	78.28	79.14	78.91	80.59	78.51
PDT	78.28	78.81	79.10	79.08	78.54

Mean over LFW, CFP-FP, AgeDB-30. EdgeFace-S, WebFace4M. Bold: best LR-trained per row.

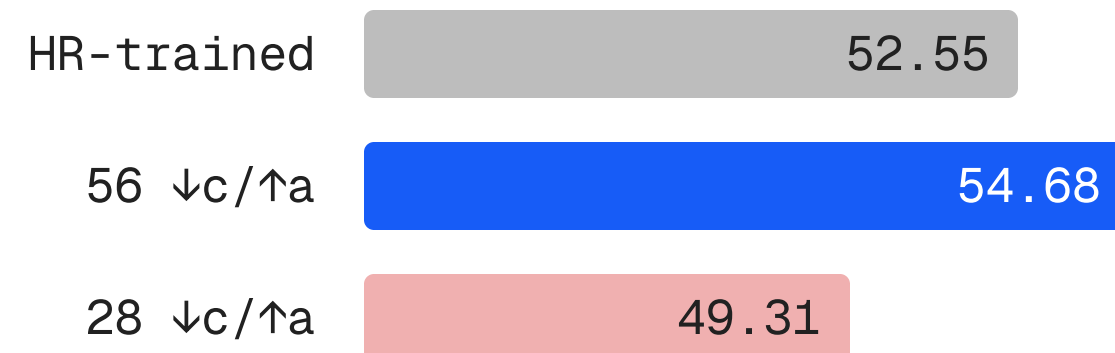
- 28 ↓c/↑a is best on **every** benchmark under HR→LR, for all three methods.
- KD gives the largest gains (+3.3 over HR-trained): the teacher supplies HR geometry the LR input cannot carry.
- If we stopped here, the recipe would be: train at 28 px, cubic down, area up, distil.

On real LR, the ranking **flips**

Synthetic HR→LR macro-mean accuracy (same EdgeFace-S models)



Real TinyFace mAP (DFA alignment)



Bar axes are truncated. 14 px not run under DFA; under aligned_pad it is also below the HR-trained model.

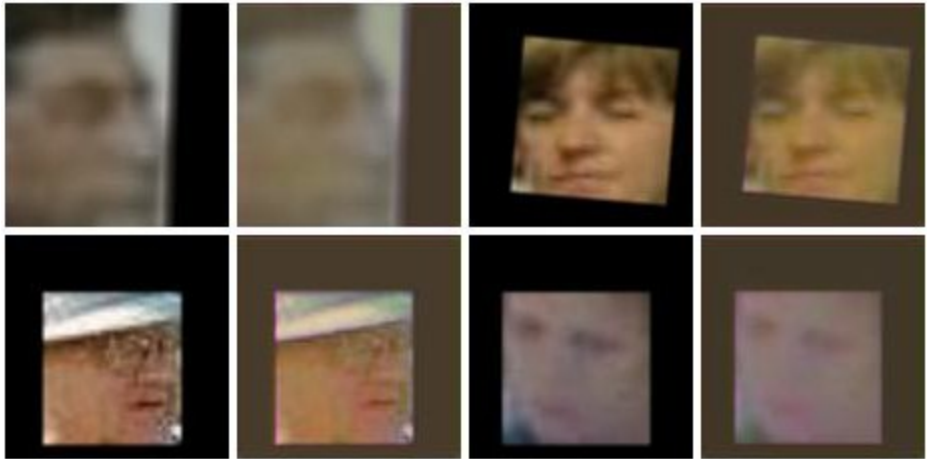
- The synthetic optimum, **28 ↓c/↑a**, is the **worst** of the family on TinyFace — below the HR-trained model.
- The milder **56 ↓c/↑a** is the best: +2.1 mAP over its own direct feed.
- Training at 56 px keeps embeddings closer to the real LR domain than a resolution tuned on synthetic data.
- Same models, same aligner. Only the evaluation data changed.

Validating on synthetic degradation alone would have picked the wrong model.

Input-side translation and distillation **do not transfer** either

TinyFace, aligned_pad	HR-tr.	56 ↓c/↑a	28 ↓a/↑c	28 ↓c/↑a	14 ↓a/↑c
LR backbone (mAP)	47.24	49.31	49.23	48.51	45.67
KD (mAP)	47.24	48.96	47.46	47.94	45.67
PDT (mAP)	47.24	46.69	46.19	45.22	43.66
LR backbone (R-1)	54.48	56.14	56.41	55.69	52.28
PDT (R-1)	54.48	53.78	53.49	52.31	51.10

Second alignment pipeline; same ordering as with DFA.



PDT outputs on native TinyFace probes (input / translated). Trained on synthetic LR, the translator mostly smooths the real probe.

- Every PDT variant is **below** the HR baseline at every resolution.
- KD, the synthetic winner, trails the plain LR backbone on real LR.

Compare each model to the **direct feed of its own backbone**

TinyFace, DFA	mAP	R-1	R-5	R-10
EdgeFace-base, frozen, WebFace12M				
base, direct feed	65.12	71.08	74.97	75.83
base + RRDB-SR + PDT	57.53	64.70	70.52	72.26
base + ESPCN-SR + PDT	56.64	63.95	69.74	71.62
EdgeFace-S, 3.65 M, WebFace4M				
S, HR-trained, direct feed	52.55	59.66	66.63	69.05
S, 56 ↓c/↑a augmentation	54.68	61.40	68.16	70.28
S, Real-ESRGAN 32 px stem	53.74	61.40	68.62	70.84
S, 56 ↓c/↑a PDT	51.94	58.88	66.17	68.51
S, 28 ↓c/↑a augmentation	49.31	57.16	64.14	66.71

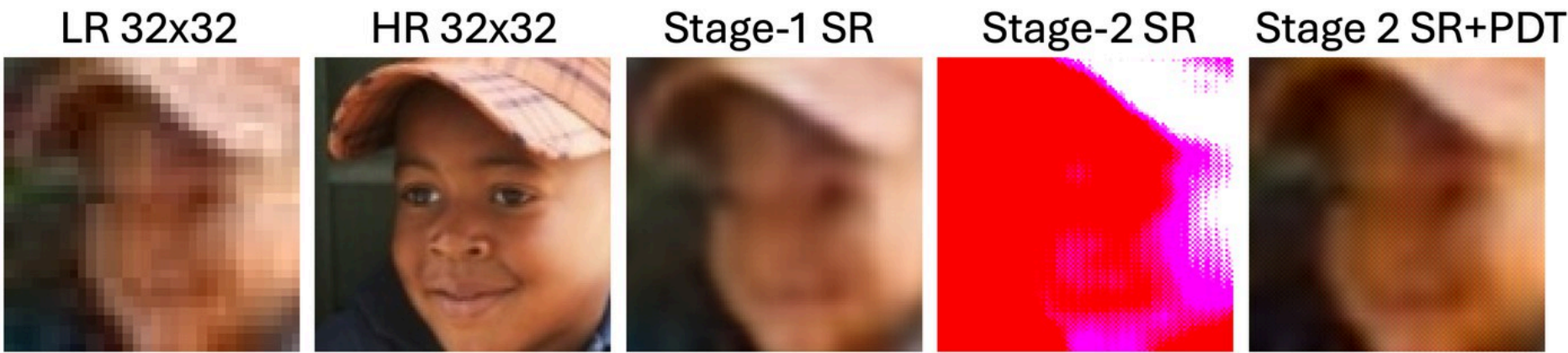
- **Compact backbone:** plain 56 px interpolation augmentation is the only synthesis that beats its own direct feed. The Real-ESRGAN stem comes close on Rank-k but not on mAP.
- **Strong frozen backbone:** the full SR + PDT pipeline sits 7.6 mAP *below* just feeding the aligned LR image.
- The learned upscaler loses to the interpolation the aligner already performs.

Plausible reconstructions, no recognition gain

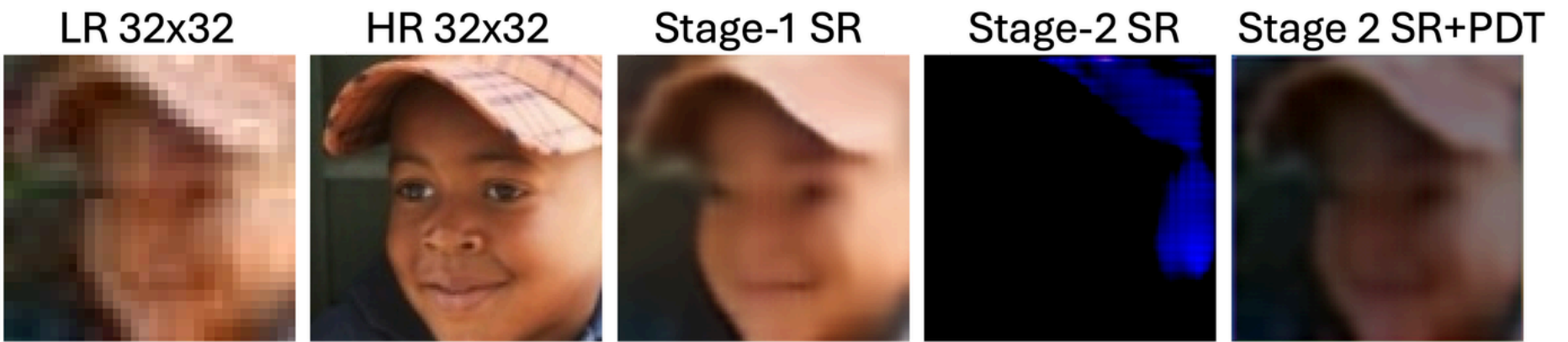
TinyFace, DFA	mAP	R-1
EdgeFace-S, direct feed	52.55	59.66
EdgeFace-S + SR only	35.82	44.02
EdgeFace-base, direct feed	65.12	71.08
EdgeFace-base + SR only	43.48	52.23
EdgeFace-base + SR + PDT	56.64	63.95

"SR only" = identity-aware ESPCN pretraining checkpoint, no joint stage.

- SR alone costs 17–22 mAP. The joint contrastive stage recovers most of it, not all.



ESPCN sub-pixel. Stage-1 output looks right; Stage-2 output is optimised for the embedding, not for a human.



RRDB (Real-ESRGAN trunk), same pattern.

Not monotonically

Strategy	Effort	Backbone	Real degr.	mAP
Direct feed, no synthesis	none	S	—	52.55
Interpolation aug, 56 ↓c/↑a	low	S	no	54.68
Interpolation aug, 28 ↓c/↑a	low	S	no	49.31
Real-ESRGAN 32 px stem	medium	S	yes	53.74
SR + PDT (ESPCN)	high	base, frozen	yes	56.64
SR + PDT (RRDB)	high	base, frozen	yes	57.53
Direct feed, base	none	base, frozen	—	65.12

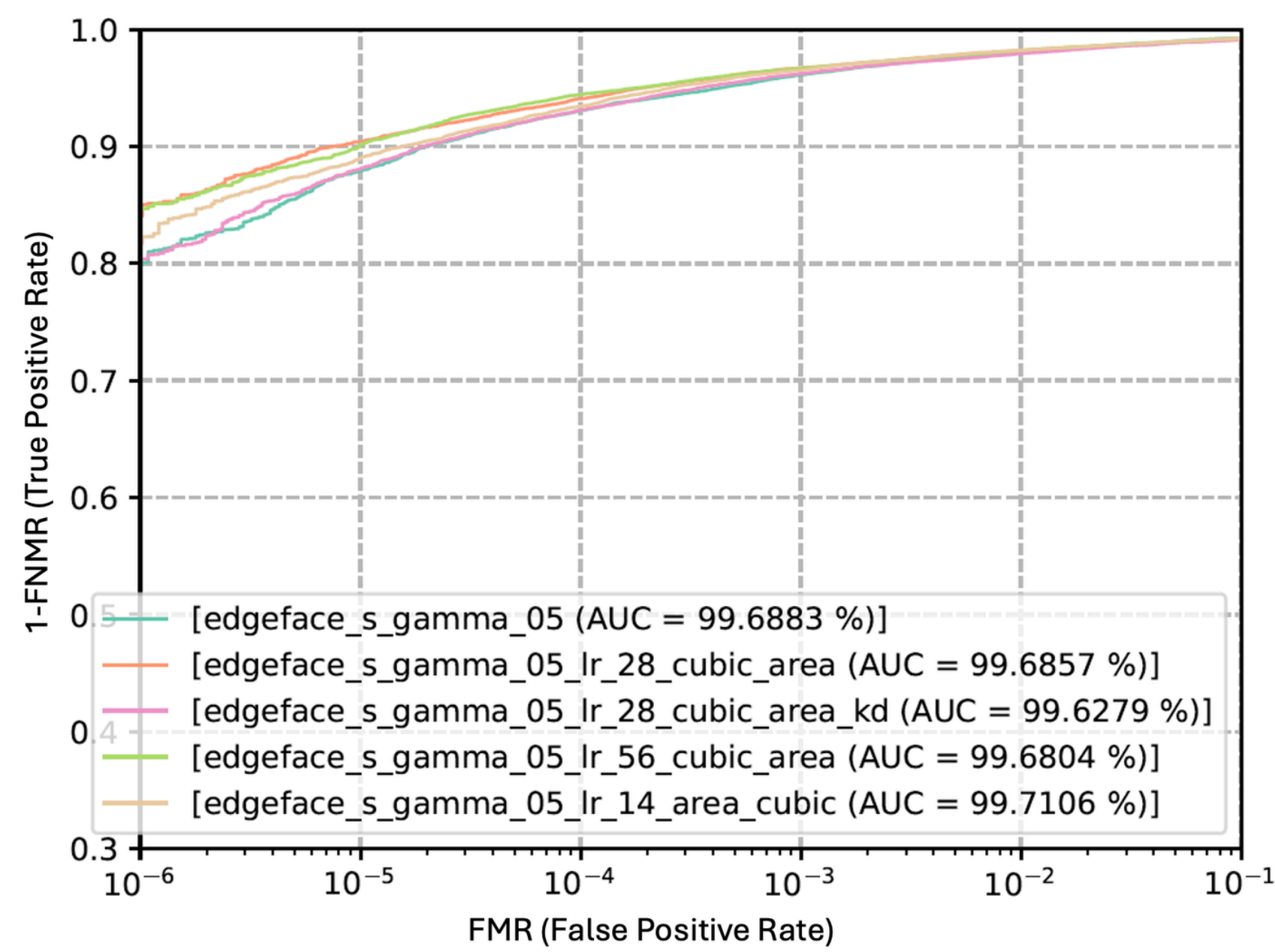
TinyFace, DFA alignment.

- The lowest-effort option on the compact, retrainable backbone is the best; the medium-effort realistic degradation does not outperform it.
- The highest-effort option is the best *adaptation* on the frozen backbone — and still loses to no adaptation at all.

LR augmentation is a **free regulariser** on HR data

HR verification	HR-tr.	56 ↓c/↑a	28 ↓a/↑c	28 ↓c/↑a
LFW	99.45	99.58	99.53	99.65
CALFW	94.53	95.25	95.28	95.32
CPLFW	90.90	91.63	91.85	91.85
CFP-FP	96.53	97.24	97.10	97.09
AgeDB-30	94.38	95.63	95.63	95.67
IJB-C TAR@10 ⁻³	96.11	96.68	96.91	96.71
IJB-C TAR@10 ⁻⁴	93.07	94.44	94.56	94.06

56 and 28 px match or beat the HR baseline on every HR set. 14 px does not; the native-32 px stem collapses on HR (LFW 93.5).



IJB-C: FMR vs. 1-FNMR. Every LR-trained EdgeFace-S variant matches or exceeds the HR baseline across operating points.

Compact state of the art on TinyFace, from training data alone

TinyFace, EFaR 2023 protocol	Params	MFLOPs	R-1	R-5
EFaR 2023 competition				
SAM-MFaceNet	1.10 M	236.8	61.31	66.33
EdgeFace-XS ($\gamma=0.6$)	1.77 M	154.0	58.77	63.89
Mod. MobileFaceNet	2.10 M	456.9	64.29	69.60
EdgeFace-S ($\gamma=0.5$), official ckpt	3.65 M	306.1	61.02	65.47
GhostFaceNetV1-1	4.09 M	215.7	59.54	64.75
Ours, same EdgeFace-S architecture, 100 epochs from scratch				
EdgeFace-S, HR-trained	3.65 M	306.1	59.66	66.63
EdgeFace-S, 56 $\downarrow$ c/ $\uparrow$ a	3.65 M	306.1	61.40	68.16

- Same architecture and budget as the official checkpoint, shorter schedule — and ahead on both Rank-1 and Rank-5.
- Mod. MobileFaceNet is more accurate at 1.5× the FLOPs.
- Our HR-trained model lands near the official checkpoint, so the protocol and alignment are comparable.

Accuracy gains **do not** reduce demographic disparity

Test px	Model	African r	Asian r	Caucasian r	Indian r	Gini
HR	HR-trained	2.80	1.01	0.25	1.39	0.37
	56 ↓c/↑a	3.36	1.62	0.10	1.82	0.36
	28 ↓c/↑a	4.24	1.85	0.05	2.47	0.39
56	HR-trained	3.41	1.92	0.15	1.03	0.41
	56 ↓c/↑a	3.25	2.04	0.06	2.51	0.33
	28 ↓c/↑a	3.85	1.28	0.10	2.06	0.41
28	HR-trained	3.36	1.34	0.17	1.33	0.42
	56 ↓c/↑a	3.92	2.17	0.11	1.08	0.45
	28 ↓c/↑a	2.59	1.76	0.14	1.62	0.31

r = group FMR / geometric-mean FMR at the IJB-C TMR@FMR=10⁻² threshold. Gini over the four group FMRs; 0 = equal.

- Caucasian FMR far below the mean, African far above, at every resolution and for every model — the known RFW ordering.
- The Gini moves in both directions with LR training; no consistent improvement.
- The largest single group ratio (4.24) belongs to the model that was best on synthetic LR.
- Below 28 px the fixed threshold stops separating scores (FNMR > 0.9); not reported.

Degradation model first, **alignment** second

DEGRADATION MISMATCH

Primary

- Interpolation does not reproduce real blur, noise and compression.
- All models share one aligner, yet the synthetic optimum is the worst on real LR — this can only be the degradation.

ALIGNMENT MISMATCH

Confounds absolute accuracy

- Landmarks are unreliable on genuinely captured LR faces.
- aligned_pad → DFA raises every model by 5–6 mAP; any TinyFace comparison must fix the aligner.

Limitations. SR front-end evaluated only on a large frozen backbone; no native-LR training data by design. Next: how much real LR each strategy needs, and learning the paired HR/LR conditional directly, evaluated on real probes.

Four things to carry out of this room

1

The degradation that is optimal on synthetic benchmarks is not optimal on real LR. 28 px wins synthetic, 56 px wins TinyFace.

3

An identity-aware SR front-end never beat feeding a strong frozen backbone the aligned LR image. Report a direct-feed baseline before claiming a restoration pipeline helps.

2

Synthesis effort does not pay off monotonically. On a compact retrainable backbone, cheap interpolation beats Real-ESRGAN degradation and a learned SR front-end.

4

Average-accuracy gains do not reduce demographic bias. RFW FMR disparity shows no consistent improvement.

Match synthesis effort to the **deployment constraint**

COMPACT MODEL YOU CAN RETRAIN

Augment mildly, retrain

- 56 px interpolation augmentation, cubic down / area up.
- +2.1 mAP on real LR, and it also improves HR and IJB-C.
- No teacher, no generator, no front-end.

STRONG BACKBONE YOU CANNOT RETRAIN

Try the direct feed first

- Align, resize, feed. It absorbs more degradation than a learned front-end recovers.
- If a front-end is still needed, train SR and PDT jointly with the recogniser loss — SR alone hurts.

Whatever you do: **validate on real native LR** with a fixed aligner, against a direct-feed baseline, and measure bias separately from accuracy.

Thank you

Code, configs and the synthesis pipeline are released.

`idiap.ch/paper/synth-lrfr`

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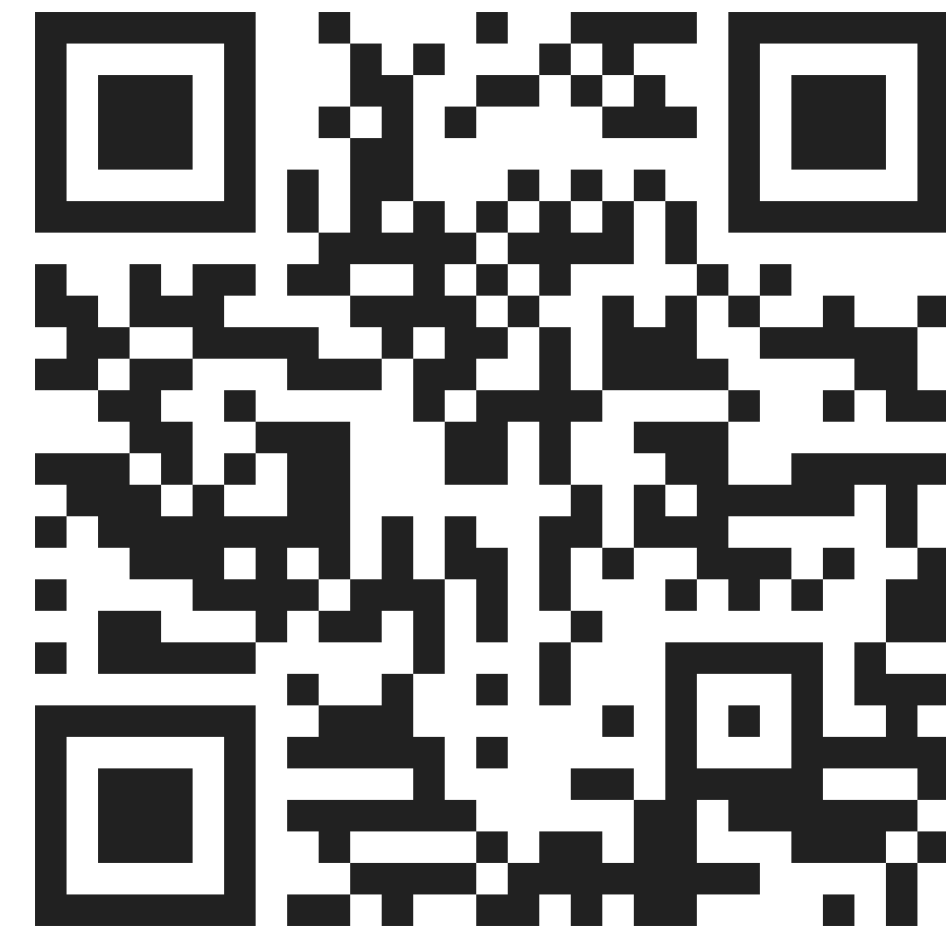
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